

CSA-CRAFT-RD-0001

**CRAFT — Collaborative Reasoning and Federated
Technical Knowledge
User and Functional Requirements Document**

Version Draft P1

Jun. 1, 2026

LiveLink

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APPROVALS

This document and all changes to it shall be approved by the appropriate Canadian Space Agency authority in accordance with CSA configuration management and document control practice. Proposed changes to the currently approved baselined version of this document shall be forwarded to the CSA Configuration Management receipt desk, or equivalent designated authority, for evaluation and submission for approval. Approved changes shall be incorporated in the next revision.

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REVISION HISTORY

Rev.	Description	Initials	Date
1.0	Initial draft.	HF	2026-05-10
1.1	Updated intern proposal structure and CRAFT scope.	HF	2026-05-21
1.2	Added use cases, definitions, and traceability matrix.	HF	2026-06-01
1.3	Added Agentic AI requirements and mapped them to use cases.	HF	2026-06-02
1.5	Added dedicated Section 4.2 Agentic AI Requirements.	HF	2026-06-02

1 INTRODUCTION

1.1 BACKGROUND

CSA systems engineering activities increasingly depend on access to distributed technical knowledge, internal standards, historical mission information, engineering analysis artefacts, risk records, and cost information. The current environment is fragmented across repositories, document stores, discipline tools, and organizational boundaries, which slows engineering work and reduces the efficiency of knowledge reuse. The CRAFT platform is intended to address that problem through a production-grade Agentic-RAG environment for engineering support, not as a pure research exercise.

CRAFT combines Large Language Model capabilities with a federated knowledge base containing CSA internal documents, standards, and historical data. It introduces autonomous AI agents able to plan, reason, retrieve information, synthesize grounded answers, support engineering workflows, and maintain auditable records of their actions.

1.2 PURPOSE

CRAFT is intended to provide CSA engineers and other authorized users with a secure and traceable platform that can query technical knowledge, support cost estimation, identify risks, detect anomalies, and generate bilingual engineering reports. This document provides the initial user and functional requirements specification for CRAFT from the viewpoint of its operational users and stakeholders.

1.3 SCOPE

This document applies to CRAFT as a production platform supporting systems engineering activities inside CSA, including document management, risk assessment, cost estimation, anomaly detection, mission planning support, and report generation. The platform is based on an Agentic-RAG architecture using autonomous agents, retrieval components, orchestration services, memory services, local and external LLM inference, and integration with CSA enterprise repositories and engineering tools.

1.4 CHANGE AUTHORITY / RESPONSIBILITY

Proposed changes to this document shall be submitted through the applicable CSA configuration management and technical authority process for consideration and disposition. The baselined version of this document shall be controlled, and approved changes shall be incorporated into the next authorized revision.

1.5 CONVENTION AND NOTIFICATION

The requirement verbs in this document shall be interpreted as follows: “shall” indicates a binding requirement that must be implemented and verified; “should” indicates a desirable goal or good practice; “may” indicates permission; “will” indicates a statement of fact or declared intent; and “is/are” indicates descriptive information. Rationales are included to provide clarification, justification, or purpose, but in the event of inconsistency, the requirement statement takes precedence over the rationale.

2 DOCUMENTS

2.1 PARENT DOCUMENTS

No parent document has been explicitly identified in the current document draft.

2.2 APPLICABLE DOCUMENTS

ID	Document Title	Reference
AD-01	CSA-SE-PR-0001	Systems Engineering Methods and Practices
AD-02	Responsible use of artificial intelligence in government	https://www.canada.ca/en/government/system/digital-government/digital-government-innovations/responsible-use-ai.html
AD-03	NASA Systems Engineering Handbook	NASA/SP-2016-6105 Rev2
AD-04	ECSS Systems Engineering Standard	ECSS-E-ST-10C
AD-05	CSA Data Classification Policy	CSA-SEC-POL-003
AD-06	Directive on Automated Decision-Making	https://www.tbs-sct.canada.ca/pol/doc-eng.aspx?id=32592
AD-07	AI governance principles	https://www.canada.ca/en/government/system/digital-government/digital-government-innovations/responsible-use-ai/principles.html
AD-08	Technology Readiness and Risk Assessment Guidelines	CSA-ST-GDL-0001

2.3 REFERENCE DOCUMENTS

ID	Document Title	Reference
RD-01	CRAFT Architecture Design Document (TBD)	TBD
RD-02	CRAFT Intern Proposal v1.2	NA
RD-03	Systems Engineering Mission Tailoring Guidelines	CSA-SE-GDL-0001

3 CRAFT CONCEPT DESCRIPTION

3.1 CRAFT OBJECTIVES

CRAFT objectives are to provide CSA with a secure and auditable platform where authorized users can discover, retrieve, analyze, and use technical knowledge and engineering data through natural language and structured workflows. The platform shall support engineering decision-making by enabling access to federated documents, technical standards, historical mission records, cost data, telemetry-derived signals, and governance-controlled AI assistance.

3.2 CONCEPT OF OPERATIONS

CRAFT will ingest documents from CSA repositories and other relevant engineering sources, extract structured content, generate embeddings, and maintain searchable vector and graph representations of the knowledge base. Authorized users will interact with the system through a user interface and APIs, using natural language queries or structured input forms for specific workflows. When a user submits a request, the platform will route that request through agent orchestration services that can decompose the task, retrieve relevant evidence, synthesize an answer, calculate confidence, and either respond directly or escalate to human review if the action or conclusion is high-risk or low confidence.

3.3 CRAFT PLATFORM USERS

CRAFT users include CSA systems engineers, [Subject Matter Experts](#), finance engineers, risk managers, operations engineers, mission control personnel, project managers, technical authorities, and governance reviewers.

User Category	Status	Definition	Notes
Admin	Implemented	Users responsible for CRAFT administration and platform support.	Corresponds to CIO team administration functions.
Employees	Implemented	Users responsible for operational, expert, and oversight use of CRAFT.	Expert users are treated as equivalent to oversight users in this version. CRAFT_SEFM_Requirements_Document_Draft_v1.3.docx
Executives	Future	Users who need executive-level access to approved outputs and summaries.	Not implemented in this version.

4 USER AND FUNCTIONAL REQUIREMENTS

4.1 FUNCTIONAL REQUIREMENTS

ID	Type	Requirement	Rationale
UMR-001	Mandatory	CRAFT shall enable engineers to query the technical knowledge base in natural language in English and French.	Supports day-to-day engineering work through direct access to federated knowledge.
UMR-002	Mandatory	CRAFT shall retrieve relevant documents and present results with source citations.	Traceable retrieval is central to trust, explainability, and engineering verification.
UMR-003	Mandatory	CRAFT shall implement a Retrieval-Augmented Generation pipeline with a minimum context window of 8192 tokens.	Provides sufficient grounded context for synthesis.
UMR-004	Mandatory	CRAFT shall return the top five ranked documents with cosine similarity score greater than or equal to 0.72.	Controls retrieval quality.
UMR-005	Mandatory	The retrieval function shall perform hybrid search combining semantic similarity and BM25 keyword relevance with a default weighting of 0.7 semantic and 0.3 keyword.	Balances semantic and lexical retrieval.
UMR-006	Mandatory	Every synthesized answer shall include traceable source references identifying document, page, and section where applicable.	Required for auditability and explainability.
UMR-007	Mandatory	CRAFT shall calculate and expose a confidence score between 0 and 1 for generated outputs.	Confidence supports escalation and review.
UMR-008	Mandatory	Outputs with confidence lower than 0.6 shall trigger escalation to a human reviewer or subject matter expert.	Low-confidence outputs shall not be treated as final engineering advice.
UMR-009	Mandatory	CRAFT shall implement a self-correction loop allowing retrieval refinement when initial evidence is insufficient.	Improves retrieval completeness before escalation.
UMR-010	Mandatory	Generated answers shall be grounded in retrieved context and shall achieve a faithfulness score of at least 0.80 using the selected evaluation framework.	Controls hallucination risk in engineering use.
UMR-011	Mandatory	CRAFT shall implement a multi-agent architecture in which each agent has one defined role and responsibility.	Role separation reduces ambiguity and supports governance.
UMR-012	Mandatory	Each agent shall operate only with its permitted tools from the tool registry.	Controls scope and unauthorized action.

Commented [CJ1]: I do not understand this column. Why don't we stick with the "shall", "should", "will" way of doing requirements?

Commented [HF2R1]: @Cusson, Jean-François (ASC/CSA) Just to help identify where can I ignore in a first phase of development. The Shall, etc. are in the Requirements column.

UMR-013	Mandatory	The agent orchestrator shall implement a ReAct loop with a maximum of 10 reasoning steps per query before escalation.	Limits uncontrolled chains and runtime cost.
UMR-014	Mandatory	CRAFT shall maintain three tiers of agent memory: episodic, semantic, and procedural.	Supports continuity, knowledge reuse, and workflow learning.
UMR-015	Mandatory	Agents shall exchange structured messages containing agent identifier, action type, input, output, confidence score, and timestamp.	Needed for orchestration control and audit.
UMR-016	Mandatory	CRAFT shall support agent spawning and termination without service interruption.	Supports availability and scaling.
UMR-017	Mandatory	Agents shall not access data or system functions outside their authorized scope.	Core data protection and governance requirement.
UMR-018	Mandatory	The planning function shall decompose complex user requests into sub-tasks before dispatching them to specialist agents.	Enables multi-step workflows.
UMR-019	Mandatory	CRAFT shall detect repetitive non-progressing agent behavior and break the loop when the same action repeats three times without progress.	Prevents runaway loops.
UMR-020	Mandatory	CRAFT shall apply guardrails that prevent off-topic, harmful, or policy-violating agent outputs.	Supports AI governance and safe operation.
UMR-021	Mandatory	CRAFT shall require Human-in-the-Loop approval for actions classified as high-risk.	Protects high-impact engineering decisions.
UMR-022	Mandatory	HITL approval requests shall display the agent reasoning chain, confidence score, and source documents used.	Reviewers need evidence to decide.
UMR-023	Mandatory	Authorized engineers shall be able to approve, reject, or modify agent recommendations before commitment.	Keeps human authority in control.
UMR-024	Mandatory	Rejected recommendations shall trigger a feedback loop allowing the agent to re-plan using the engineer correction.	Supports corrective iteration.
UMR-025	Mandatory	HITL approval requests shall time out after 24 hours if no response is received, and the requested action shall be cancelled and flagged for review.	Controls latent pending actions.
UMR-026	Mandatory	All HITL decisions shall be logged with engineer identifier, decision, and justification text.	Approval traceability is required.
UMR-027	Mandatory	CRAFT shall maintain an immutable audit log of agent actions and approval decisions.	Log integrity is required for governance.
UMR-028	Mandatory	Agent model weights and prompt templates shall be version controlled and reviewed before deployment.	Supports controlled configuration.

UMR-029	Mandatory	Where technically feasible, agent actions shall be reversible or support rollback.	Limits damage from incorrect actions.
UMR-030	Mandatory	CRAFT shall apply rate limiting per agent, with a default maximum of 100 LLM calls per agent per minute.	Protects runtime stability.
UMR-031	Mandatory	CRAFT shall generate mission cost estimates from historical data.	Supports engineering and finance planning.
UMR-032	Mandatory	Cost similarity matching shall use a weighted normalized Euclidean distance method.	Defines the estimation approach.
UMR-033	Mandatory	The platform shall retrieve the five most similar historical missions for a candidate estimate.	Provides evidence for estimates.
UMR-034	Mandatory	Cost estimation outputs shall include confidence information and the source missions used in the estimate.	Supports reviewer validation.
UMR-035	Mandatory	Any estimate associated with a financial commitment or baseline decision shall require Human-in-the-Loop approval before use.	Protects high-risk commitments.
UMR-036	Goal	CRAFT shall identify and flag risks in mission documents.	<i>Need to specify the risk standards supporting proactive engineering risk management.</i>
UMR-037	Mandatory	The risk function shall use keyword pattern matching combined with LLM-based classification.	Defines the risk identification method.
UMR-038	Mandatory	Identified risks shall be classified by severity.	Supports triage and governance.
UMR-039	Mandatory	Proposed updates to the risk register shall require Human-in-the-Loop approval before commitment.	Protects controlled risk records.
UMR-040	Mandatory	Risk outputs shall remain grounded in the document evidence from which they were derived.	Controls hallucination in risk classification.
UMR-041	Mandatory	CRAFT shall detect anomalies in telemetry and system data.	Supports operational monitoring.
UMR-042	Mandatory	The anomaly detection function shall use a Z-score threshold greater than 2.5 unless otherwise configured.	Defines the default anomaly threshold.
UMR-043	Mandatory	The anomaly agent shall retrieve similar historical anomaly cases from the knowledge base.	Improves diagnosis and context.
UMR-044	Mandatory	High-severity anomalies shall trigger Human-in-the-Loop acknowledgement or review.	Protects operations decisions.
UMR-045	Mandatory	All anomaly events and user responses shall be logged in the audit trail.	Maintains event traceability.
UMR-046	Mandatory	CRAFT shall generate bilingual reports using approved templates.	Supports CSA reporting requirements.

Commented [CJ3]: There must be a way to specify a standard format for the risk reporting?

Commented [HF4R3]: @Cusson, Jean-François (ASC/CSA) Absolutely. I postponed the requirement.

Commented [CJ5]: Isn't this out of scope, and too broad?

Commented [HF6R5]: @Cusson, Jean-François (ASC/CSA) Agreed. Deleted.

UMR-047	Mandatory	The reporting function shall compile data from approved internal sources including databases, registers, and retrieved documents.	Ensures reports use controlled sources.
UMR-048	Mandatory	Narrative report sections shall be grounded in retrieved evidence and identified source data.	Prevents fabricated report content.
UMR-049	Mandatory	Report drafts shall support engineer review before finalization.	Maintains human review authority.
UMR-050	Mandatory	Final reports shall be versioned and logged.	Supports document control.
UMR-051	Mandatory	CRAFT shall integrate with existing CSA document repositories.	Core operational integration need.
UMR-052	Mandatory	CRAFT shall connect to SharePoint and Confluence through supported APIs.	Implements repository integration.
UMR-053	Mandatory	CRAFT shall support integration with CSA financial systems such as SAP for cost-related workflows where authorized.	Supports cost workflows.
UMR-054	Mandatory	CRAFT shall support integration with mission design and analysis tools such as STK and MATLAB where operationally required.	Supports engineering workflow continuity.
UMR-055	Goal	CRAFT should expose approved functions through REST and GraphQL interfaces to support controlled interoperability.	Improves interoperability.
UMR-056	Mandatory	CRAFT shall respond to user queries in less than 5 seconds for 90 percent of queries under nominal operating conditions.	Defines performance target.
UMR-057	Mandatory	CRAFT shall provide 99.5 percent availability during CSA business hours.	Defines availability target.
UMR-058	Mandatory	CRAFT shall protect sensitive CSA data in accordance with the applicable CSA data classification policy for the deployment scope.	Protects sensitive information.
UMR-059	Mandatory	CRAFT shall support at least 500 concurrent users as additional directorates adopt the platform.	Defines scalability target.
UMR-060	Mandatory	Every final user-facing answer shall include source references sufficient to explain the basis of the output.	Defines explainability requirement.
UMR-061	Mandatory	CRAFT shall generate and maintain decision traceability reports that record the evidence sources consulted, reasoning outputs produced, confidence levels assigned, and human approvals or rejections associated with recommendations and decisions.	Supports auditability, explainability, and reconstruction of the decision-making process. It defines when reporting is required (medium/high-risk decisions) and what

			information must be captured.
UMR-062	Mandatory	CRAFT shall provide authorized users with the ability to retrieve and export audit reports containing a time-stamped history of significant system activities, including user requests, agent actions, data sources accessed, recommendations generated, decisions taken, and subsequent modifications. Supports governance, compliance reviews, lessons learned activities, and post-decision investigations.	Supports UMR-061 and implement the reporting capabilities. It defines who can access the reports (authorized users), what outputs are expected (exportable reports), and ties retention to an external governing policy rather than an arbitrary duration.

4.2 AGENTIC AI REQUIREMENTS

This section defines the specific requirements applicable to the Agentic AI functions of CRAFT. These requirements apply to autonomous agents, agent orchestration, agent memory, reasoning control, Human-in-the-Loop review mechanisms, and the safety and governance controls required for operational use within CSA. In the event of inconsistency between this section and supporting use cases or rationale text, the requirement statements in this section shall take precedence.

4.2.1 AGENT ARCHITECTURE REQUIREMENTS

The following requirements define the baseline architectural constraints for agent roles, orchestration behavior, memory structure, message exchange, lifecycle management, and permission boundaries.

Tag	Nature of the Requirement	Requirement	Rationale
AAR-001	Mandatory	CRAFT shall implement a multi-agent architecture in which each agent has one defined role and responsibility.	Clear role separation reduces ambiguity, improves control, and supports verification of agent behavior.
AAR-002	Mandatory	Each agent shall operate with a defined set of permitted tools from the approved Tool Registry, and no agent shall access tools outside its permitted set.	Tool restriction is required to prevent unauthorized action and reduce operational risk.
AAR-003	Mandatory	The Agent Orchestrator shall implement a Reasoning and Acting workflow with a maximum of 10 reasoning steps per user request before escalation or termination.	A bounded reasoning cycle is required to control runtime cost, failure propagation, and runaway execution.
AAR-004	Mandatory	CRAFT shall implement agent memory using three tiers: episodic memory for current-session context, semantic memory for persistent knowledge, and procedural memory for learned workflows.	Tiered memory supports continuity, retrieval efficiency, and controlled reuse of prior operational knowledge.
AAR-005	Mandatory	Agents shall communicate using a structured message format containing, at a minimum, agent identifier, action type,	Structured messaging is required for orchestration,

		input, output, confidence score, and timestamp.	traceability, and auditability.
AAR-006	Mandatory	CRAFT shall support agent spawning and termination without interruption to the availability of the platform.	Agent lifecycle flexibility is required for scaling and operational resilience.
AAR-007	Mandatory	CRAFT shall enforce agent permission boundaries such that no agent may modify system configuration or access data outside its authorized scope.	Permission boundaries are required to protect system integrity and sensitive CSA information.

4.2.2 AGENT REASONING REQUIREMENTS

The following requirements define how CRAFT agents shall decompose tasks, retrieve evidence, generate grounded outputs, calculate confidence, and refine retrieval when the initial context is insufficient.

Tag	Nature of the Requirement	Requirement	Rationale
ARR-001	Mandatory	The planning function shall decompose complex user requests into sub-tasks before dispatch to specialist agents.	Task decomposition is required for multi-step workflows and controlled specialization.
ARR-002	Mandatory	The retrieval function shall perform hybrid search combining semantic similarity and BM25 keyword relevance, with a default weighting of 0.7 semantic and 0.3 keyword unless otherwise configured.	Hybrid retrieval improves robustness across natural language and domain-specific terminology.
ARR-003	Mandatory	The synthesis function shall cite every source used in a generated answer by document identifier and, where available, page number and section reference.	Source traceability is required for explainability and engineering verification.
ARR-004	Mandatory	Each agent shall calculate and report a confidence score between 0 and 1 for its output.	Confidence scoring supports review, escalation, and operational trust.
ARR-005	Mandatory	Outputs with confidence lower than 0.6 shall trigger escalation to a human reviewer or subject matter expert.	Low-confidence outputs shall not be treated as authoritative operational guidance.
ARR-006	Mandatory	CRAFT shall implement a self-correction loop permitting agents to re-query or refine parameters when the initial retrieval does not provide sufficient context.	Refinement is required to improve evidence completeness before escalation.
ARR-007	Mandatory	All generated content shall be grounded in retrieved context and shall achieve a faithfulness score of at least 0.80 using the approved evaluation framework.	Grounding and faithfulness controls are required to limit hallucination risk.

4.2.3 HUMAN-IN-THE-LOOP REQUIREMENTS

The following requirements define the controls by which authorized CSA personnel shall review, approve, reject, or modify agent-generated recommendations before high-risk actions are committed.

Tag	Nature of the Requirement	Requirement	Rationale
HITL-001	Mandatory	CRAFT shall implement mandatory Human-in-the-Loop approval gates for actions classified as high-risk.	High-impact actions require explicit human authority before execution or commitment.
HITL-002	Mandatory	High-risk actions shall include, at a minimum, cost commitments, risk register updates, and mission parameter changes.	These actions can materially affect engineering, operational, or governance outcomes.
HITL-003	Mandatory	HITL approval requests shall display the reasoning chain, confidence score, and source documents used by the agent.	Reviewers require visibility into the basis of the recommendation to make an informed decision.
HITL-004	Mandatory	Authorized engineers shall be able to approve, reject, or modify any agent recommendation before it is committed to the system.	Human reviewers must retain final authority over controlled actions.
HITL-005	Mandatory	Rejected agent recommendations shall trigger a feedback loop in which the agent re-plans using the reviewer correction where such correction is provided.	This supports correction, learning, and controlled iteration.
HITL-006	Mandatory	HITL approval requests shall time out after 24 hours if no disposition is provided, and the requested action shall be cancelled and flagged for review.	Timeout behaviour is required to prevent indefinite pending high-risk actions.
HITL-007	Mandatory	All HITL decisions, including approval, rejection, and modification, shall be logged with reviewer identifier, decision, timestamp, and justification text.	Full decision traceability is required for governance and audit.

4.2.4 AGENTIC AI SAFETY AND GOVERNANCE REQUIREMENTS

The following requirements define the safety, control, audit, rollback, rate-limiting, and configuration-management obligations applicable to the CRAFT agentic AI environment.

Tag	Nature of the Requirement	Requirement	Rationale
ASG-001	Mandatory	CRAFT shall implement agent guardrails that prevent agents from generating content that is off-topic, harmful, or in violation of CSA data policy.	Guardrails are required for safe and policy-compliant operational use.

ASG-002	Mandatory	Where technically feasible, all agent actions shall be reversible or shall support rollback capability.	Reversibility limits the impact of incorrect or premature actions.
ASG-003	Mandatory	CRAFT shall maintain an immutable audit log of all agent decisions and agent actions, and audit entries shall not be modifiable or deletable through normal system operations.	Immutable audit logging is required for governance, incident review, and compliance.
ASG-004	Mandatory	CRAFT shall implement rate limiting per agent, with a default maximum of 100 LLM calls per agent per minute unless otherwise approved and configured.	Rate limiting is required to control resource consumption and prevent runaway execution.
ASG-005	Mandatory	CRAFT shall detect repetitive non-progressing agent behaviour and shall break the loop if the same action is repeated three times without progress, after which the workflow shall be escalated or terminated.	Loop detection is required to prevent ineffective or unsafe repeated execution.
ASG-006	Mandatory	Agent model weights, prompt templates, and equivalent operational inference configurations shall be version controlled, reviewed, and approved before deployment to production.	Configuration control is required to maintain reproducibility, traceability, and governance discipline.

APPENDIX A — USER ROLE DEFINITIONS, OPERATIONAL ASSUMPTIONS, RISK AND MITIGATION, AND INITIAL ON-PREMISES INFRASTRUCTURE ROM

Purpose. This appendix is provided as a standalone draft appendix for the CRAFT requirements package. It defines the current user role structure, records operational assumptions, summarizes key risks and mitigations, and provides a rough-order-of-magnitude (ROM) estimate for an initial on-premises deployment. The ROM values are indicative planning figures only and shall not be treated as final design values. Final architecture, hardware counts, storage capacity, GPU allocation, resiliency design, and network sizing shall be defined and controlled in the CRAFT Architecture Design Document (RD-01).

A.1 User Role Definitions

User Category	Status in Current Version	Definition	Primary Responsibilities	Access Scope
Admin	Implemented	Administrative users responsible for platform administration and support, corresponding to CIO team functions or equivalent authorized administrators.	User and permission administration; configuration management; monitoring; logging review; integration support; operational continuity.	Administrative access limited to approved platform administration functions and subject to audit and security control.
Employees	Implemented	Authorized CSA personnel using CRAFT for operational, analytical, engineering, and oversight activities. This category includes both operational users and expert users; for this version, expert users are treated as equivalent to oversight users.	Query technical knowledge; review retrieved evidence; perform engineering analysis; review and disposition recommendations where authorized.	Role-based access to approved tools, documents, and review functions in accordance with need-to-know and security controls.
Executives	Not implemented in current version	Senior stakeholders requiring executive-level summaries, dashboards, or approved outputs for governance and decision support.	Review approved outputs and summary information for decision support.	Not in scope for the current version; to be defined in a future revision.

A.2 Operational Assumptions

Assumption ID	Operational Assumption	Basis
OA-01	The current implementation baseline supports Admin and Employees only; Executive user functions are deferred to a future revision.	Section 3.3 revision direction and current system scope.
OA-02	CRAFT is intended to operate as a production platform for CSA engineering support rather than as a research prototype.	Section 1 purpose and scope.
OA-03	The platform target availability is 99.5 percent during CSA business hours.	UR-NF02 and KPI-06.
OA-04	This standalone Appendix A uses an initial planning assumption of up to 50 concurrent users for first deployment sizing. This does not supersede the formal scalability requirement in the main requirements document.	Initial deployment ROM planning assumption; main document UR-NF04 remains separate until formally revised.
OA-05	The platform processes information at the deployment tier stated in the current draft baseline and enforces data protection controls accordingly.	UR-NF03 and related governance controls.
OA-06	Human-in-the-Loop approval remains mandatory for high-risk actions.	SR-F08 and Section 4.2.3.
OA-07	Workflow-specific user categorization is excluded from the current version.	Requested Section 3.3 change.
OA-08	The platform interfaces with existing enterprise systems including SharePoint, Confluence, SAP, STK, and MATLAB.	Section 2.3 system boundary.
OA_09	Capability to generate a report providing details of the process leading to a decision (audit).	UMR-031 and UMR-032 (New requirements.)

Commented [CJ7]: Capability to generate a report providing details of the process leading to a decision (audit)

Commented [HF8R7]: @Cusson, Jean-François (ASC/CSA) Agreed. Added OA-09.

A.3 Risk and Mitigation

Risk ID	Risk Description	Probability	Impact	Mitigation
AR-01	LLM-generated technical outputs may be incorrect or insufficiently grounded.	Medium	High	Enforce grounding and faithfulness controls, source citation, and confidence-based escalation.
AR-02	Agents may attempt actions outside their authorized scope.	Low	High	Enforce tool permission boundaries, role restrictions, and immutable audit logging.
AR-03	Runaway loops or excessive inference calls may consume	Low	Medium	Apply per-agent rate limiting, loop detection, and escalation or

	local infrastructure resources.			termination rules.
AR-04	Human review queues may delay operational decisions.	Medium	Medium	Apply timeout handling, escalation rules, and role-based review routing.
AR-05	Sensitive information may be exposed through retrieval or response generation.	Low	High	Enforce data classification controls, agent guardrails, access control, and auditability.
AR-06	Confidence scoring may be poorly calibrated, causing over-trust or excessive escalation.	Medium	Medium	Perform calibration testing and maintain held-out evaluation sets.
AR-07	On-premises infrastructure may be undersized for peak concurrent use.	Medium	High	Apply staged rollout, load testing, and capacity margin before operational rollout.

A.4 Initial On-Premises Infrastructure Rough Order of Magnitude

Important note. The following sizing values are preliminary ROM planning figures intended to support budgeting, space, and procurement discussion only. They are not final design commitments. The definitive server topology, redundancy model, storage architecture, model hosting strategy, and network design shall be defined in RD-01 CRAFT Architecture Design Document after corpus sizing, benchmark testing, and security architecture review.

Sizing Item	Initial ROM Proposal for 50 Concurrent Users	Comment
Application and API servers	1 to 2 servers	Supports UI, API gateway, tool registry, and orchestration functions; 2 servers if redundancy is desired.
LLM/GPU inference servers	1 GPU server initially; 2 if local redundancy is required	Initial local inference pool for agent workloads and priority queries.
Database and vector store servers	1 primary server plus backup or replica plan; 2 if HA is required	PostgreSQL and pgvector initial deployment sized for reduced concurrency.
Graph database server	1 server	Initial Neo4j deployment; may be co-hosted in pilot conditions if workload remains light.

Memory and cache server	1 server	Redis or equivalent for session and short-term memory functions; may be co-hosted initially if acceptable.
Monitoring and logging server	1 server	Prometheus, Grafana, central logging, and audit collection.
Ingestion and parsing worker servers	1 server	Supports document ingestion, parsing, ETL, and background processing.
Management or backup server	0 to 1 server	May be implemented as a separate server or consolidated into administrative infrastructure.
Total initial server count	Approximately 1 to 2 server roles	Depends on consolidation approach, HA posture, and virtualization platform.

A.5 ROM Storage, GPU, and Network Sizing

Category	Initial ROM Proposal for 50 Concurrent Users	Comment
Primary high-performance storage	10 to 15 TB usable	For indexed documents, vector data, graph data, active reports, databases, and high-use working sets.
Archive or lower-cost storage	20 to 40 TB usable	For historical documents, backups, snapshots, and lower-access content.
Backup retention storage	15 to 30 TB usable	Depends on retention policy, snapshot frequency, and replication strategy.
GPU allocation	1 to 2 enterprise GPUs initially	Final quantity depends on chosen local model size, concurrency, and latency targets.
System memory	256 GB to 512 GB aggregate RAM across core servers	Supports databases, caching, orchestration, parsing, and concurrent user sessions.
Core network uplinks	Dual 1 Gbps	Suitable initial baseline for east-west traffic across storage, inference, and database services; lower speeds may be acceptable only for pilot or non-performance-critical components.
Storage network	1 Gbps baseline	Provides performance margin for indexing, retrieval, and backup windows without

		over-sizing the initial deployment.
External or enterprise connectivity (TBD)	Redundant enterprise links sized to synchronization and user access demand	Exact WAN sizing depends on SharePoint, Confluence, SAP, and external repository traffic patterns.

A.6 Basis for the ROM Estimate

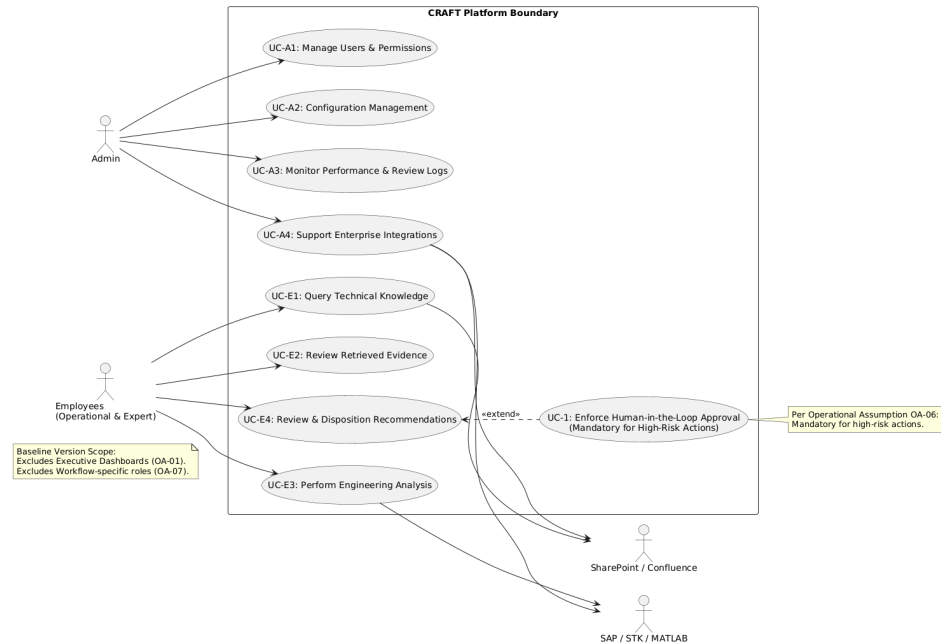
The ROM estimate is derived from the current draft baseline requiring support for an Agentic-RAG architecture including ingestion, parsing, embeddings, vector storage, graph storage, orchestration, inference, memory, API, UI, and monitoring functions, while this standalone appendix uses an initial deployment planning assumption of up to 50 concurrent users.

The estimate also assumes an initial on-premises deployment aligned with CSA control of enterprise data, auditability, and integration to internal repositories and engineering tools. It includes modest resilience and capacity margin, but it is intentionally conservative and incomplete for design approval purposes.

A.7 Design Authority Statement

This appendix does not supersede the CRAFT Architecture Design Document. All final implementation details, including exact hardware bill of materials, virtualization layout, clustering, backup design, security zoning, disaster recovery, storage classes, and network topology, shall be specified and approved in RD-01 CRAFT Architecture Design Document before procurement or deployment.

APPENDIX B — OPERATIONAL USE CASES



For each use case, provide an editable table covering: Use Case ID, Actor, Pre-conditions, Post-conditions, Main Success Scenario (Step-by-step), and Exception Flows.

Use Case 1: Systems Engineering Documentation Ingestion & Reasoning

- UC-A1: Ingestion and Concept Deduction: Parse incoming space engineering PDFs into a unified Knowledge Graph (Neo4j). Deduced new concepts, identifies technical contradictions or design deviations against previously ingested legacy CSA baseline files.

Use Case 2: Global Space Tech Benchmarking & Horizon Scanning

- UC-B1: Cross-Sector Technology Comparison: Run automated web-scrapes and open-access paper ingestion pipelines. Benchmark global space tech against CSA guidelines, Federal/Provincial published needs, and Canadian university/private sector research (covering Rovers, Space Robotics, Space Debris, Earth Observation, etc.).

Use Case 3: Space Engineering Costing Workflow

Develop the detailed engineering flows for these 5 exact sub-use cases:

- UC-C1: Dublin Core Metadata Extraction: Scan unstructured PDF directories, extract metadata from Page 1 (Title, Company, Doc Number, Date) via local LLM vision/text extraction, and format as Dublin Core schema.

2. UC-C2: Verifiable QA with Deep Citations: Answer user queries (e.g., “what is the mass of the instrument?”) providing the exact text answer along with an explicit hyperlink mapping to the exact localized file path and page number.
3. UC-C3: Tabular Data Transformation to SQL: Parse complex PDF tables (RAM, Risk Register, Mass, Power Consumption, TRL) into structured formats using open-source extraction tools (e.g., Tabula, Camelot, or LlamaParse) and write to a PostgreSQL database.
4. UC-C4: Parametric Mission Distance Calculation: Calculate mathematical nearest neighbors using a weighted Mahalanobis or Normalized Euclidean parametric distance algorithm across historical space missions.

The system must compute the distance $D(x, y)$ between a target mission x and an archive mission y using the following formula to handle features across different engineering scales (e.g., Mass in kg vs. Power in Watts):

$$D(x, y) = \sqrt{\sum_{i=1}^n w_i \left(\frac{x_i - y_i}{\sigma_i} \right)^2}$$

Where:

- x_i, y_i : The value of parameter i (e.g., Mass, Power, TRL, Payload Volume) for the target and archive missions.
 - σ_i : The standard deviation of parameter i across the entire historical CSA mission database (for normalization).
 - w_i : The user-defined or agent-assigned engineering weight for parameter i , where $\sum w_i = 1$.
5. UC-C5: Automated Report Compilation: Draft a comprehensive Microsoft Word (.docx) cost report via a Python backend (e.g., python-docx), using an approved institutional template and dynamically injecting data queried directly from the local PostgreSQL database.

UC-C1 Natural Language Technical Query

Actor: CSA Engineer

Description: The user asks a technical question in natural language. CRAFT retrieves relevant information from the CSA knowledge base and returns a grounded answer with citations and confidence information.

Pre-conditions:

The engineer is authenticated in CRAFT.

The knowledge base is indexed with current CSA documents.

Main Success Scenario:

1. The engineer submits a technical question in English or French.
2. The planning function decomposes the request into retrieval sub-tasks.
3. The retrieval function performs hybrid search across the indexed knowledge sources.
4. The synthesis function generates a grounded answer using retrieved evidence.
5. CRAFT presents the answer with source citations and confidence information.
6. The engineer reviews the answer and supporting sources.

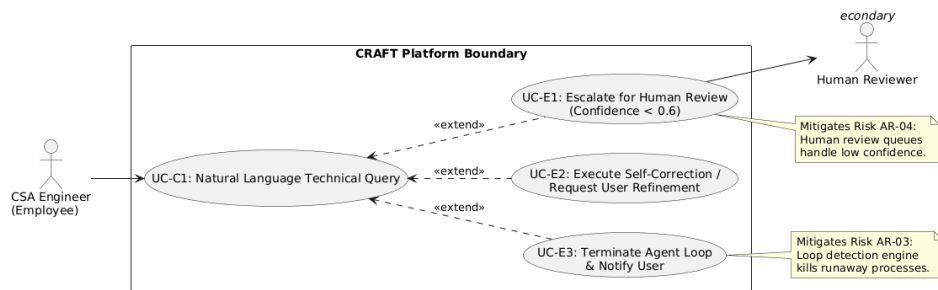
Exception Flows:

E1. If confidence is lower than 0.6, the result is escalated for human review.

E2. If relevant evidence is insufficient, CRAFT requests refinement or performs a self-correction loop.

E3. If agent looping is detected, the loop is broken and the user is notified.

Applicable Requirement ID	Application in Use Case
UMR-001	Natural language query entry point.
UMR-002	Retrieval with cited sources.
UMR-005	Hybrid search implementation.
UMR-006	Mandatory citations in answer.
UMR-007	Confidence score shown to user.
UMR-008	Escalation on low confidence.
UMR-009	Retrieval self-correction when context is insufficient.
UMR-010	Grounding and faithfulness of response.
UMR-011	Role-based multi-agent support.
UMR-013	Bounded ReAct reasoning loop.
UMR-019	Loop detection and break.



UC-C2 Cost Estimation ~~from Historical Data~~

Actor: CSA Finance Engineer, Systems Engineer

Description: The user provides mission parameters. CRAFT searches historical mission cost data and generates a cost estimate by analogy using weighted normalized Euclidean distance similarity matching; as well as parametric estimation using COTS tools like True Planning, SSCM, PCEC, NICM.

Pre-conditions:

Historical mission cost data are available in the approved database.

COTS tools are available and can interface with Excel

The cost estimation workflow and finance tools are operational.

Main Success Scenario:

1. The engineer submits a mission concept document and Product Breakdown structure

2. The planning function creates a Cost Breakdown Structure suitable for parametric analysis

3. The engineer submits mission parameters such as mass, orbit, duration, heritage, lifetime, and complexity.

4. The planning function decomposes the request into a structured cost feature vector (computing the parametric distance).

5. The cost estimation function computes similarity against historical missions.

4. CRAFT retrieves the closest historical mission analogues.

5. CRAFT generates a cost estimate by analogy with confidence information and supporting references.

6. CRAFT proposes analysis scenarios for sensitivity analysis using parametric estimation.

7. CRAFT generates a cost estimate using parametric estimation with confidence information and supporting references.

8. A Human-in-the-Loop approval request is issued for high-risk or commitment-related use.

9. The engineer approves, rejects, or modifies the proposed estimate.

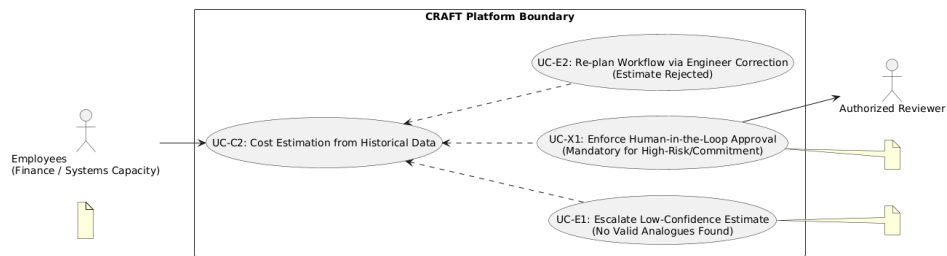
10. Approved estimates are logged in the audit trail.

Exception Flows:

E1. If no acceptable analogue missions are found, the estimate is flagged as low confidence and escalated.

E2. If the estimate is rejected, the workflow re-plans using the engineer correction.

Applicable Requirement ID	Application in Use Case
UMR-021	HITL for high-risk actions.
UMR-022	Reasoning chain shown during approval.
UMR-023	Approve, reject, or modify action.
UMR-024	Feedback loop after rejection.
UMR-026	HITL decision logging.
UMR-027	Immutable audit trail.
UMR-031	Mission cost estimation capability.
UMR-032	Weighted normalized Euclidean distance method.
UMR-033	Use of five similar historical missions.
UMR-034	Confidence and cited evidence in estimate.
UMR-035	Approval required before commitment.



UC-C3 Risk Identification and Flagging

Actor: Systems Engineer, Risk Manager

Description: The user requests risk analysis on a mission document or document set. CRAFT identifies candidate risks, classifies them by severity, and proposes updates to the risk register subject to approval.

Pre-conditions:

The mission document is available in CRAFT or is uploaded by an authorized user.

Risk taxonomy and classification criteria are available.

The risk register is available for controlled update.

Main Success Scenario:

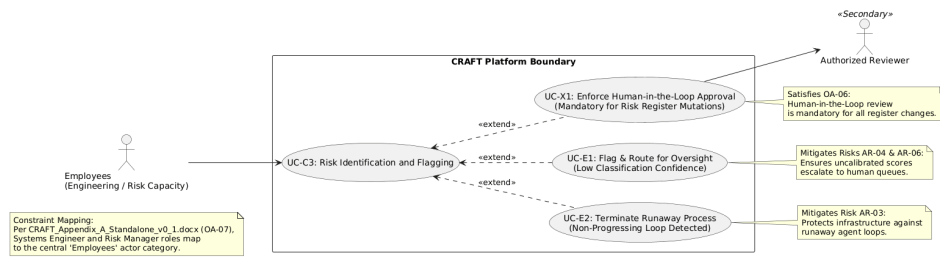
1. The user requests risk analysis on the selected document set.
2. The planning function creates a risk analysis workflow.
3. The risk function scans the content using keyword pattern matching and LLM classification.
4. CRAFT identifies candidate risks and classifies them by severity.
5. CRAFT proposes updates to the risk register.
6. A Human-in-the-Loop approval request is issued before any risk register change is committed.
7. The engineer or risk manager approves, rejects, or modifies the proposed updates.
8. Approved actions are written to the audit trail.

Exception Flows:

E1. If classification confidence is low, the result is flagged as uncertain and mandatory human review applies.

E2. If repeated non-progressing risk flagging occurs, the loop is broken and escalated.

Applicable Requirement ID	Application in Use Case
UMR-017	Permission boundaries protect unrelated systems.
UMR-019	Loop detection for repetitive risk flagging.
UMR-021	Approval required for high-risk controlled actions.
UMR-022	Reasoning chain shown to reviewer.
UMR-023	Reviewer can approve, reject, or modify.
UMR-026	Decision logging.
UMR-027	Immutable audit trail.
UMR-036	Risk identification capability.
UMR-037	Keyword plus LLM classification method.
UMR-038	Severity classification.
UMR-039	Risk register changes require approval.
UMR-040	Risk outputs grounded in source evidence.



UC-C4 Anomaly Detection in System Data

Actor: Operations Engineer, Mission Control

Description: CRAFT monitors telemetry and system data, detects anomalies using a statistical threshold, retrieves similar historical cases, and notifies operations personnel with confidence and context.

Pre-conditions:

Telemetry data streams are connected to CRAFT ingestion services.

Thresholds and notification rules are configured.

Authorized operations users are defined.

Main Success Scenario:

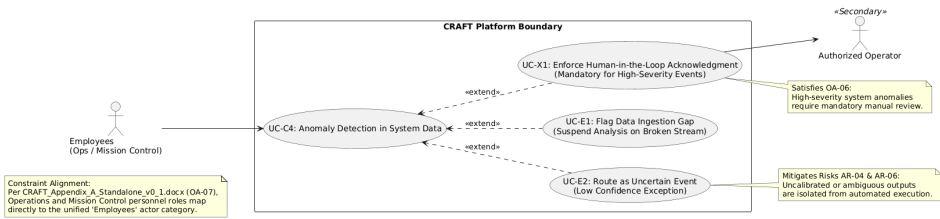
1. CRAFT monitors incoming telemetry and system data.
2. The anomaly function computes Z-scores against historical baselines.
3. When the configured threshold is exceeded, the anomaly is flagged.
4. CRAFT retrieves similar historical anomaly cases from the knowledge base.
5. CRAFT generates an anomaly report with likely causes, evidence, and confidence information.
6. High-severity anomalies trigger Human-in-the-Loop acknowledgement or review.
7. Operations personnel acknowledge, investigate, or dismiss the anomaly.
8. All anomaly events and decisions are logged.

Exception Flows:

E1. If the data stream is interrupted, CRAFT flags a data gap and pauses anomaly analysis for that stream.

E2. If confidence is low, the anomaly is marked uncertain and routed for human review without automated action.

Applicable Requirement ID	Application in Use Case
UMR-014	Use of semantic and episodic memory for baselines and session context.
UMR-016	Persistent background anomaly agent lifecycle support.
UMR-019	Loop control for repeated anomaly behavior.
UMR-021	HITL on high-severity anomalies.
UMR-022	Reasoning chain shown in review.
UMR-025	Timeout behavior for unresolved approvals.
UMR-027	Immutable logging.
UMR-030	Rate limiting for anomaly agents.
UMR-041	Anomaly detection capability.
UMR-042	Default Z-score threshold.
UMR-043	Retrieval of similar historical anomaly cases.
UMR-044	Human review for high-severity anomalies.
UMR-045	Event and response logging.



UC-C5 Automated Report Generation

Actor: Systems Engineer, Project Manager

Description: The user requests a structured report. CRAFT compiles data from controlled sources, generates a bilingual report draft using approved templates, and routes the draft for review before finalization.

Pre-conditions:

- Approved report templates are available.
- Required data sources are accessible.
- Authorized report users are defined.

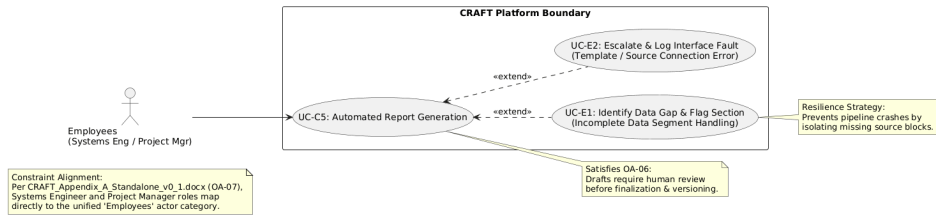
Main Success Scenario:

1. The user specifies report type, scope, and period.
2. The planning function decomposes the request into data collection and synthesis tasks.
3. CRAFT queries approved structured and unstructured sources.
4. CRAFT generates narrative sections grounded in retrieved evidence.
5. CRAFT inserts the content into the approved bilingual template.
6. A reviewable draft is presented to the engineer or project manager.
7. The reviewer approves, edits, or requests regeneration.
8. The final report is versioned and logged.

Exception Flows:

- E1. If data are incomplete, CRAFT identifies the gap and flags the affected section for review.
- E2. If the template or source interface fails, CRAFT escalates the issue with an explicit error record.

Applicable Requirement ID	Application in Use Case
UMR-022	Reviewer sees supporting reasoning and sources.
UMR-023	Approve, reject, or modify draft outcome.
UMR-027	Immutable audit trail.
UMR-028	Version-controlled prompts and generation controls.
UMR-029	Draft rollback or discard before commitment.
UMR-046	Bilingual report generation capability.
UMR-047	Compilation from approved internal sources.
UMR-048	Narrative grounded in evidence.
UMR-049	Engineer review before finalization.
UMR-050	Versioning and logging of final reports.



UC-C6 Enterprise Cognitive Synthesis Layer for Aerospace Engineering Lifecycle Management

Actor: Systems Engineer, Cost Analyst, Scientist, Data Governance Officer

Description: The user submits a domain-specific technical query against the CRAFT federated knowledge corpus. The platform executes multi-turn contextual reasoning over validated institutional data, enforces governance constraints, and returns a traceable, bilingual response entirely within the secure perimeter.

Pre-conditions:

- Federated data repositories are classified, ingested, and accessible.
- The SEFM is deployed and validated for domain-specific inference.
- Role-based access control policies are defined per directorate.
- Containerized infrastructure is provisioned and hardened.

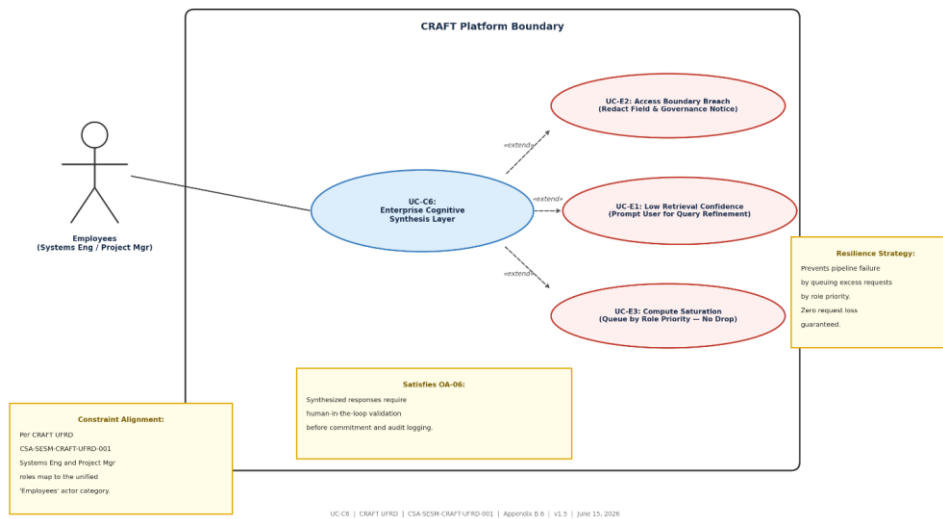
Main Success Scenario:

1. The authenticated user submits a natural language query via the bilingual integration endpoint.
2. The semantic layer parses the query and routes it to the appropriate knowledge partition.
3. The Hybrid Retrieval Engine executes parallel semantic vector search and explicit relationship graph traversal.
4. Retrieved context is ranked and passed to the SEFM inference pipeline.
5. The SEFM executes multi-turn reasoning and synthesizes a response with full source traceability.
6. The guardrail compliance layer validates the output against access entitlements and governance constraints.
7. The validated response is returned to the user with inline traceability references.
8. The transaction is logged to the container-isolated audit trail with full lineage metadata.

Exception Flows:

- E1. If retrieval confidence falls below threshold, CRAFT prompts the user for query refinement before re-executing.
- E2. If the output breaches a directorate access boundary, the response is redacted at field level and a governance notice is issued.
- E3. If the inference stack reaches capacity under high concurrency, requests are queued by role priority and no requests are dropped.

Applicable Requirement ID	Application in Use Case
UMR-022	Reviewer sees supporting reasoning and sources.
UMR-023	Approve, reject, or modify draft outcome.
UMR-027	Immutable audit trail.
UMR-028	Version-controlled prompts and generation controls.
UMR-029	Draft rollback or discard before commitment.
UMR-046	Bilingual report generation capability.
UMR-047	Compilation from approved internal sources.
UMR-048	Narrative grounded in evidence.
UMR-049	Engineer review before finalization.
UMR-050	Versioning and logging of final reports.



APPENDIX C — TRACEABILITY TABLES

C.1 User Requirements to Prior Source Requirements

Legacy Requirement ID	Mapped UMR IDs
UR-F01	UMR-001, UMR-003, UMR-005, UMR-009
UR-F02	UMR-002, UMR-006, UMR-010, UMR-060
UR-F03	UMR-031, UMR-032, UMR-033, UMR-034, UMR-035
UR-F04	UMR-036, UMR-037, UMR-038, UMR-039, UMR-040
UR-F05	UMR-041, UMR-042, UMR-043, UMR-044, UMR-045
UR-F06	UMR-011, UMR-013, UMR-014, UMR-018
UR-F07	UMR-015, UMR-026, UMR-027, UMR-029
UR-F08	UMR-021, UMR-022, UMR-023, UMR-024, UMR-025
UR-F09	UMR-046, UMR-049, UMR-050
UR-F10	UMR-051, UMR-052
UR-NF01	UMR-056
UR-NF02	UMR-016, UMR-025, UMR-030, UMR-057
UR-NF03	UMR-017, UMR-020, UMR-058
UR-NF04	UMR-016, UMR-059
UR-NF05	UMR-006, UMR-007, UMR-022, UMR-060

C.2 Functional Requirement to Use Case Mapping

UMR ID	UC-C1	UC-C2	UC-C3	UC-C4	UC-C5
UMR-001	✓				
UMR-002	✓				
UMR-005	✓			✓	

UMR-006	✓	✓			✓
UMR-007	✓	✓	✓	✓	
UMR-008	✓	✓	✓	✓	
UMR-009	✓				
UMR-010	✓		✓	✓	✓
UMR-011	✓	✓	✓	✓	✓
UMR-013	✓		✓		
UMR-014				✓	
UMR-015		✓			✓
UMR-016				✓	
UMR-017			✓		
UMR-018	✓	✓	✓		✓
UMR-019	✓		✓	✓	
UMR-020	✓		✓		
UMR-021		✓	✓	✓	
UMR-022		✓	✓	✓	✓
UMR-023		✓	✓		✓
UMR-024		✓			
UMR-025				✓	
UMR-026		✓	✓		
UMR-027		✓	✓	✓	✓
UMR-028					✓
UMR-029		✓			✓
UMR-030				✓	
UMR-031		✓			
UMR-032		✓			
UMR-033		✓			
UMR-034		✓			
UMR-035		✓			
UMR-036			✓		
UMR-037			✓		
UMR-038			✓		
UMR-039			✓		
UMR-040			✓		
UMR-041				✓	
UMR-042				✓	
UMR-043				✓	
UMR-044				✓	
UMR-045				✓	
UMR-046					✓
UMR-047					✓
UMR-048					✓

UMR-049					✓
UMR-050					✓
UMR-051	✓				
UMR-052	✓				
UMR-053		✓			
UMR-054					
UMR-055	✓				✓
UMR-056	✓				
UMR-057	✓	✓	✓	✓	✓
UMR-058	✓	✓	✓	✓	✓
UMR-059	✓	✓	✓	✓	✓
UMR-060	✓				✓

APPENDIX D — ACRONYMS

Acronym	Definition
AI	Artificial Intelligence
API	Application Programming Interface
CoT	Chain-of-Thought
CRAFT	Collaborative Reasoning and Federated Technical Knowledge
CSA	Canadian Space Agency
ECSS	European Cooperation for Space Standardization
HITL	Human-in-the-Loop
LLM	Large Language Model
RAG	Retrieval-Augmented Generation
RAGAS	RAG Assessment evaluation framework
ReAct	Reasoning and Acting
REST	Representational State Transfer
SAP	Systems, Applications, and Products
SESM	Space Engineering and Systems Management
STK	Satellite Tool Kit
UMR	User and Functional Requirement
UFRD	User and Functional Requirements Document